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1 Free fall

1.1 Drop times

Each group dropped a steel ball from five heights and timed the fall with a photogate. The acceleration follows from the height and the time of the fall.

Table 1: Drop times for five heights.

Height (m)	Time (s)	g (m/s^2)
0.50	0.319	9.83
0.75	0.391	9.81
1.00	0.452	9.79
1.25	0.505	9.80
1.50	0.553	9.81

The values agree within half a per cent, which is less than the uncertainty of a single timing.

1.2 Quiz scores

Table 2: Frequency of quiz scores.

Score	Students	Share
6	3	0.15
7	6	0.30
8	7	0.35
9	4	0.20

2 Tables without rules

2.1 Values of a cubic

The table below gives $f(x) = x^3 - 2x$ at evenly spaced points. It is set without any rules, as many instructors type tables in a hurry.

x	$f(x)$	$f'(x)$	Difference
-2	-4	10	5.0
-1	1	1	-1.0
0	0	-2	-1.0
1	-1	1	5.0
2	4	10	17.0

Table 3: Values of f and its derivative.

2.2 Heat of materials

Table 4: Specific heat by phase.

Type	Material	Specific heat (J/kg K)	
		Solid	Liquid
Metal	Aluminium	900	1180
	Copper	385	530
Other	Water	2090	4186
	Ethanol	2420	2440

3 Quantities and a worksheet

3.1 Definitions

Table 5: Mechanical quantities.

Quantity	Symbol	Definition
Momentum	$\vec{p}$	$\vec{p} = m\vec{v}$
Kinetic energy	K	$K = \frac{1}{2}mv^2$
Work	W	$W = \vec{F} \cdot \vec{d}$
Power	P	$P = W/t$

3.2 Record your data

Fill in one row for each trial during the lab.

Table 6: Pendulum trials.

Trial	Length (m)	Ten swings (s)	Period (s)

4 Not tables

4.1 Key terms

Displacement The change in position of an object, measured as a straight line from where it started to where it ended, with a direction.

Velocity How fast the position changes and in which direction; the slope of a position against time graph at any instant.

Acceleration How fast the velocity changes; it can point against the motion, in which case the object slows down.

4.2 Matrices and systems

The rotation by a quarter turn and the system it solves are

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 4 \end{pmatrix}, \qquad \left[\begin{array}{ccc|c} 2 & 1 & 0 & 3 \\ 1 & 3 & 1 & 5 \\ 0 & 1 & 4 & 5 \end{array}\right].$$

Elimination gives

$$\begin{aligned} 2x + y &= 3, \\ x + 3y + z &= 5, \\ y + 4z &= 5, \end{aligned}$$

and the sign function is

$$\operatorname{sgn}(x) = \begin{cases} -1 & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ 1 & \text{if } x > 0. \end{cases}$$